

TOPLITA, Romania

WMO: 151070

Lat: 46.9264N

Lon: 25.3600E

Elev: 2258

StdP: 13.54

Time Zone: 2.00 (EUE)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.7	1.4	-4.9	4.6	-1.6	-0.7	5.7	3.5	20.6	20.3	16.7	22.6	0.9	130	0.558

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.2	85.4	65.8	82.4	64.6	79.3	63.7	67.8	80.8	66.4	78.7	65.0	76.8	7.4	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.6	95.7	71.8	62.1	90.8	70.6	60.9	86.7	69.0	33.5	81.0	32.3	78.7	31.2	76.8	75.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.4	13.8	11.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.6	22.2	26.8	35.4	45.1	54.7	61.7	63.7	64.2	54.5	45.2	35.0	25.8		
	DBStd	16.46	9.22	9.02	7.66	7.83	5.68	5.39	5.08	5.45	6.82	6.87	8.41	9.41		
	HDD50	3583	861	651	451	186	21	1	0	0	33	176	452	751		
	HDD65	7568	1326	1071	916	598	319	127	85	80	317	612	901	1216		
	CDD50	1625	0	0	0	38	168	352	426	440	170	29	2	0		
	CDD65	135	0	0	0	0	1	29	46	55	4	0	0	0		
	CDH74	2226	0	0	0	16	74	447	643	899	146	1	0	0		
	CDH80	566	0	0	0	0	8	100	167	267	24	0	0	0		
(14) Wind	WSAvg	3.8	3.0	3.2	4.1	4.9	4.4	4.5	4.5	3.9	3.7	3.3	2.7	2.9		
(15) Precipitation	PrecAvg	27.6	1.1	1.3	1.5	2.2	3.1	4.0	3.8	3.0	2.4	2.1	1.7	1.7		
	PrecMax	38.2	2.1	2.7	3.7	3.7	5.7	7.2	6.8	4.8	5.8	4.2	4.0	4.8		
	PrecMin	18.9	0.4	0.2	0.3	0.9	1.2	1.3	1.3	1.3	0.5	0.0	0.0	0.5		
	PrecStd	4.6	0.5	0.6	0.8	0.9	1.1	1.4	1.5	1.0	1.3	1.1	0.9	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.8	50.3	65.5	76.1	80.7	86.9	88.4	90.4	82.8	72.2	59.2	46.9		
		MCWB	38.2	43.6	50.4	55.7	60.0	67.2	66.9	66.4	62.4	56.9	51.2	43.1		
	2%	DB	38.0	44.2	58.3	71.8	75.7	82.8	84.4	85.9	77.7	68.1	53.6	41.7		
		MCWB	35.8	40.2	46.9	54.5	59.6	65.5	65.9	65.2	61.0	55.0	48.7	40.0		
	5%	DB	35.3	40.4	52.3	67.2	72.1	79.2	80.9	82.7	73.5	63.1	49.6	38.4		
		MCWB	33.7	37.1	43.4	52.6	58.4	64.2	65.1	64.7	59.8	53.0	45.8	37.0		
	10%	DB	33.7	37.4	47.0	61.3	68.3	75.1	77.2	79.1	68.1	57.6	45.9	35.9		
		MCWB	32.5	34.8	40.6	49.9	56.2	62.0	63.9	63.8	57.8	51.1	43.4	34.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.0	45.1	51.6	57.2	62.9	69.5	69.2	69.8	64.7	59.0	52.0	43.9		
		MCDB	41.0	50.2	63.9	72.7	74.9	83.4	82.0	84.5	78.1	68.4	57.6	45.6		
	2%	WB	35.9	40.9	47.7	55.5	60.9	67.2	67.4	67.7	62.6	56.4	49.3	39.8		
		MCDB	37.4	43.8	57.4	69.7	72.6	79.7	80.4	80.1	75.1	65.3	53.7	41.3		
	5%	WB	34.1	37.8	44.5	53.4	59.3	65.3	66.1	66.2	60.8	54.3	46.7	37.4		
		MCDB	35.1	40.2	52.0	66.1	70.5	76.9	77.9	78.9	72.0	61.6	49.3	38.6		
	10%	WB	32.6	35.2	41.4	51.0	57.5	63.5	64.7	64.8	58.8	51.6	43.9	34.9		
		MCDB	33.6	36.6	46.1	60.2	66.3	74.0	75.3	77.2	67.4	56.9	46.1	35.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	13.7	16.4	19.7	23.2	23.2	23.9	24.2	26.8	22.8	22.3	14.3	11.4		
		MCDBR	11.7	17.3	30.6	35.8	30.3	31.2	31.6	33.9	32.3	33.2	20.1	12.2		
		MCWBR	9.0	12.8	19.0	18.7	15.1	13.4	14.1	14.4	15.8	20.0	14.1	10.1		
	5% WB	MCDBR	10.2	16.4	29.7	32.2	27.8	28.5	27.4	29.3	29.6	30.1	18.0	12.4		
		MCWBR	8.3	12.2	18.4	16.9	14.4	13.1	12.7	13.4	14.8	18.2	12.6	10.2		
(40) Clear-Sky Solar Irradiance	taub		0.302	0.334	0.382	0.443	0.419	0.413	0.437	0.444	0.399	0.360	0.332	0.305		
	taud		2.303	2.225	2.174	2.064	2.211	2.265	2.201	2.177	2.268	2.366	2.378	2.348		
	Ebn at Noon		246	260	263	257	269	271	262	254	255	249	233	231		
	Edh at Noon		24	32	40	49	44	42	44	43	36	28	23	21		
(44) All-Sky Solar Radiation	RadAvg		397	664	1013	1355	1670	1794	1781	1642	1171	798	449	303		
	RadStd		56	75	125	135	160	150	158	149	140	103	60	39		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (63 neighbors)		+1.26	N/A	N/A	+1.53	N/A	N/A	-284	-396	+149	+34

Nomenclature: See separate page